Kaylee Y. Vo

Curriculum Vitae

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Google Scholar Profile

Research Interests. Scientific machine learning, which includes physics-informed neural networks and neural operators. Interest extends to AI safety, focusing on mechanistic interpretability and causal steering of language models.

Education

2024–2026 **Master of Liberal Arts in Extension Studies, Data Science**, *Harvard University*, Cambridge, MA, GPA: 4.0/4.0
Fields: Scientific Machine Learning, Applied Mathematics
Capstone: "Trajectory-Aware Prompt Optimization for AI Agents"

2012–2016 **Bachelor of Arts, Economics**, *Stanford University*, Stanford, CA, GPA: 3.4/4.0
Minor: Statistics

Research Experience

August 2026 - Present **Research Data Scientist**, *Sandia National Laboratories*, Albuquerque, NM
Project 1: Pretraining neural operators for topology optimization on geometric brackets.
Project 2: Generative modeling of kinematics using diffusion models and neural operators.

June 2026 - Aug 2026 **Research Intern**, *EleutherAI*, Seattle, WA
Advisor: Goncalo Paulo, PhD
Project: Investigating the Stability of Natural Language Autoencoders

2025–2026 **Research Associate**, *Harvard John A. Paulson School of Engineering and Applied Sciences*, Cambridge, MA
Advisor: Prof. Pavlos Protopapas
Project: Variational Boosting for Physics-Informed Neural Networks

2016–2017 **Research Associate**, *Federal Reserve Bank of San Francisco*, San Francisco, CA
Advisor: Dr. Adam Shapiro
Projects: Using Brexit to Identify the Nature of Price Rigidities; estimating FOMC/market sentiment from meeting transcripts using NLP

2015–2016 **Research Fellow**, *Stanford Center on Poverty and Inequality*, Stanford, CA
Advisors: Prof. Mark Duggan, Prof. David Rogosa, Dr. Charles Varner
Project: The Effect of Prescription Drug Monitoring Programs on the Opioid Epidemic

Awards, Fellowships, and Patents

2026 **Harvard University, Division of Continuing Education, Dean's List Academic Achievement Award**

2019 **Patent #11562425**: System for automated origination of capital based on point-of-sale data informed by location

2019 **Patent #11100575**: System for automated origination of capital based on point-of-sale data informed by time of year

2019 **Patent #10956974**: Dynamic origination of capital pricing determination based on forecasted point-of-sale revenue

2012–2016 **Stanford University Academic Scholarship** (full-ride)

Fall 2015 **Stanford Economics Top Student Award** (3 consecutive months)

Summer 2015 **Stanford Undergraduate Grant for Economics Research**

Summer 2015 **Stanford Haas Center for Public Service Research Grant**

2015–2016 **Undergraduate Research Fellow Award**, Center on Poverty and Inequality at Stanford University

Professional Experience

Mar 2023–May 2024 **Software Engineer, Data Engineering**, *Nuna Inc.*, San Francisco, CA

- Deployed risk-adjustment model (CDPS) to EMR Spark clusters, using Prefect for task orchestration. Optimized the CDPS algorithm, resulting in an 80% reduction in runtime.
- Built data transformations to map external client data to Nuna's data schema. Schemas are defined using Avro files and written to an AWS Glue. Performed Spark optimizations to reduce the ETL runtime. The work culminated in a Glue database used across the company.

Nov 2022–Mar 2023 **Senior Data Scientist**, *Carbon Health*, San Francisco, CA

- Built DBT transformations on top of Snowflake to procure data for Looker dashboards. Dashboards were used to measure provider efficiency, with metrics such as risk-adjusted total spending per patient per month.

Nov 2021–Nov 2022 **Data Scientist**, *Pngme*

- Developed an MVP loan default prediction model. The model was a logistic regression fitted on cross-sectional data. Features are financial metrics transformed into weight-of-evidence features using the OptBinning library. Lift, profit curves, and Shapley values were used as monitoring metrics, and AUC was the performance metric.
- Led feature engineering and modeling R&D work to ensure ML model performance increases commensurately with data. We generated custom word embeddings with BERT using users' SMS data as input. The final model was a mixture of experts, with a logistic regression meta-model. The model was used by several banks in Nigeria, Ghana, and Kenya.

Nov 2018–Apr 2020 **Data Scientist**, *Toast Inc.*, Boston, MA

- Built a new line of business called Toast Capital. The product consists of three patented models, an ETL data pipeline, and API endpoints. Total origination was over $15 million, the second most profitable line of business.
- Designed prototypes to improve the PD model. This includes model selection, model stability, model parsimony, and testing regression assumptions. Also led an R&D initiative to develop a generalized PD model.

Jun 2017–Oct 2018 **Data Scientist**, *Moody's Analytics*, New York, NY

- Computer Vision: Developed an in-house OCR software using Tesseract and OpenCV to parse text and objects from images. OCR extracted text, tables, and numbers from PDF's, augmenting Moody's training data, increasing the PD model's accuracy.
- Sentiment Analysis: Constructed models to determine if sentiment scores improve the predictive power of Moody's default probability model. Utilized a variety of specifications, namely Kalman Filters, VARs, and MIDAS regressions.

Teaching Experience

Jan 2026 Graduate Teaching Fellow, Mathematical Modeling (Applied Math 115) - Harvard College

Publications

Under Review K. Vo, P. Protopapas. "Variational Boosting for Physics-Informed Neural Networks." Submitted to *AISTATS*, 2026. arXiv:2607.23940

Abstract *We introduce a variational boosting framework where solutions are constructed additively in function space. Each stage minimizes the nonlinear residual over an affine subspace of a Hilbert space. Because each correction network is small, the minimization admits second-order updates.*

Presentations

Oct 2026 (scheduled) "Variational Boosting for Physics-Informed Neural Networks" *(oral and poster)*. Women in Statistics and Data Science, American Statistical Association.

Sept 2026 "Variational Boosting for Physics-Informed Neural Networks" *(oral)*. International Conference on Scientific Computing and Machine Learning (SCML 2026), University of Bath, U.K.

Aug 2026 "Trajectory-Aware Prompt Optimization for AI Agents" *(poster)*. Agentic AI Summit, UC Berkeley Center for Responsible, Decentralized Intelligence.

Jul–Aug 2026 "Variational Boosting for Physics-Informed Neural Networks" *(poster)*. STAI-X, American Statistical Association.

Nov 2016 "Measuring News Sentiment from FOMC Transcripts" *(invited talk)*. City College of San Francisco; featured on Bloomberg News.

Jun 2016 "The Effect of Prescription Drug Monitoring Programs on the Opioid Epidemic" *(poster)*. Stanford Undergraduate Research Symposium.

Skills

Programming Python (scikit-learn, PyTorch, TensorFlow, JAX), R, SQL (Postgres, Snowflake, BigQuery), Bash, MATLAB

ML & AI LLMs (LLaMA, BERT), Diffusion Models (U-Net, Stable Diffusion), Computer Vision (R-CNN, ViT, segmentation), Multimodal (CLIP)

Mathematics	Stochastic & Partial Differential Equations, Real & Functional Analysis, Matrix Theory, Causal Inference, Time Series
Infrastructure	AWS (SageMaker, S3, EC2, EMR, Glue), Kubernetes, Spark, Docker, Prefect, Git/GitHub Actions
Languages	English (native), Korean (TOPIK I - Level 2), Vietnamese (limited proficiency)

References

Available upon request.